

62. Which among the following is correctly matched?

- (a) Manganese – Bailadila
 (b) Bauxite – Lohardaga
 (c) Iron – Koderma
 (d) Mica – Talcher

R.A.S./R.T.S. (Pre) 2021

Ans. (b)

Bauxite deposits are found in Lohardaga (Jharkhand), while Koderma is famous for mica, Talcher for coal and Bailadila iron ore.

63. Match list - I with list -II and select the correct answer using the code given below :

List - I (Minerals)	List - II (Industries in which largely used)
A. Limestone	1. Cement
B. Copper	2. Electrical goods
C. Bauxite	3. Aeroplane Manufacturing
D. Manganese	4. Steel

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	3	1	4
(c)	4	3	2	1
(d)	3	4	1	2

U.P.R.O./A.R.O. (Pre) 2021

Ans. (a)

(Minerals)	(Industries in which largely used)
Limestone	– Cement
Copper	– Electrical goods
Bauxite	– Aeroplane Manufacturing
Manganese	– Steel

64. The Central Government has announced the opening of first lithium mine of India in which of the following districts of Chhattisgarh?

- (a) Bastar
 (b) Raigarh
 (c) Korba
 (d) Janjgir-Champa

Chhattisgarh P.C.S. (Pre) 2024

Ans. (c)

Lithium reserves by the Geological Survey of India (GSI). The Central Government decided to open the first lithium mine in the Katghora region of Korba district, Chhattisgarh in August 2024.

This announcement was made during the sixth Governing Body meeting of the National Mineral Exploration Trust in New Delhi. The GSI reported lithium concentrations ranging from 10 to 2,000 parts per million (PPM) over an area of approximately 250 hectares, with the possibility of even higher concentration.

Electrical Energy

(i) Thermal

*Vital energy like coal, crude oil and natural gas are used to produce thermal electrical energy in India. The **National Thermal Power Corporation (NTPC)** was established in 1975. As per the Ministry of Power, the installed power capacity in India as of December 31, 2024 was 462001.66 MW. In this, thermal energy has the highest share of **52.89%** i.e. 244376.91 MW. Coal contributes 212349.50 MW, lignite 6620.00 MW, gas 24818.21 MW and diesel 589.20 MW to thermal power installed capacity. The share of other sectors in the total power generation of India is: **hydropower** 46968.17 MW, nuclear power 8180 MW, and **non-conventional energy (renewable energy)** 162476.58 MW. *Government of India is going to start an **Ultra Mega Power Project** based on coal with Power Finance Corporation as its nodal agency. Usage of super critical technique is proposed in this project to cut the carbon emissions. Each of these Ultra Mega Power Projects has a capacity of **4,000 MW** or above. *Tata Power is establishing an Ultra Mega Power Project in **Mundra** (Gujarat). According to Ministry of Power, initially, following nine (9) numbers Ultra Mega Power Projects (UMPPs) were proposed to be set up in different states:

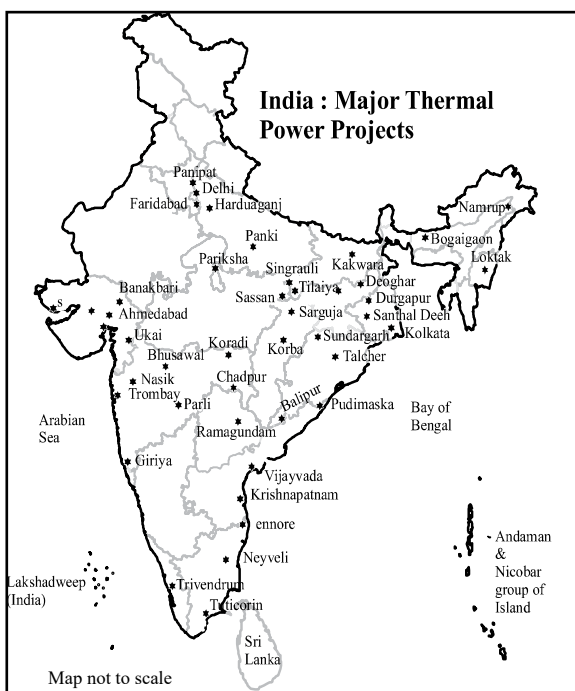
- Sasan UMPP in Madhya Pradesh - Coal pithead,
- Mundra UMPP in Gujarat - Coastal,
- Krishnapatnam UMPP in Andhra Pradesh - Coastal,
- Tilaiya UMPP in Jharkhand - Coal pithead
- UMPP in Chhattisgarh - Coal pithead,
- Bedabahal UMPP in Odisha - Coal pithead,
- Cheyur UMPP in Tamil Nadu - Domestic coal based,
- UMPP in Maharashtra - Coastal,
- UMPP in Karnataka - Coastal.

In addition to nine UMPPs originally identified a request has come from some of the state governments for installation of additional UMPPs in their states. These are given below-

- (i) Two additional UMPPs in Odisha
- (ii) 2nd UMPP in Gujarat
- (iii) 2nd UMPP in Jharkhand - Deoghar UMPP
- (iv) 2nd UMPP in Tamil Nadu
- (v) UMPP in Bihar - Bank UMPP
- (vi) UMPP in Uttar Pradesh
- (vii) 2nd UMPP in Andhra Pradesh.

Neyveli Thermal Power Corporation in Tamil Nadu is the first plant of South-East Asia working on lignite coal. Lignite is a tertiary category of coal. ***NTPC Farakka Thermal Power Plant** is situated in Murshidabad district of West Bengal. It gets water from Farakka feeder canal and coal from Rajmahal coalfield. **Ramagundam Super Thermal Power Station** is situated in Telangana. Its total capacity is 2600 MW.

***Obra Thermal Power Station** was established in 1967 with the help of Russian engineers in Sonbhadra district of Uttar Pradesh. ***Bokaro Thermal Power Plant** is situated in Jharkhand.



*Major Thermal Power Plants of India include : **Ukai Thermal Power Corporation** (Gujarat), **Patratu Thermal Power Plant** (Jharkhand), **Pench Thermal Power Plant**

(Madhya Pradesh), **Dabhol Thermal Power Plant** (Maharashtra), **Utran Thermal Power Plant** (Gujarat), **Raichur** (Karnataka), etc. **Kothagudem Thermal Power Plants** situated in Telangana. **Uran Gas Turbine Power Station** is a gas based power project situated in Raigad district of Maharashtra. It is operated by Maharashtra State Power Generation Company Limited.

1. With reference to NTPC unit at Singrauli, which of the following statement/s is / are correct ?

1. It is situated in northern part of Chhattisgarh State.
2. It is NTPC's oldest unit.

Select the correct answer using the codes given below:

Codes :

- | | |
|------------------|---------------------|
| (a) Only 1 | (b) Only 2 |
| (c) Both 1 and 2 | (d) Neither 1 nor 2 |

U.P.P.C.S. (Pre) 2021

Ans. (b)

The thermal power unit of NTPC at Singrauli was established in the year 1977, while it has been since the year 1982. This is the first unit of NTPC. Singrauli region is spread over 2 districts of 2 states i.e. Sidhi in Madhya Pradesh and Sonbhadra in Uttar Pradesh. There are 3 super thermal power projects in this area. Singrauli and Rihand are in Sonbhadra district of Uttar Pradesh and Vindhyachal Plant is in Singrauli district of Madhya Pradesh.

2. Which of the following pairs are correctly matched?

- | | |
|---------------|----------------------------|
| 1. Idukki | : Thermal power station |
| 2. Sabarigiri | : Hydro - electric project |
| 3. Ghatprabha | : Irrigation Project |
| 4. Ramganga | : Multipurpose project |

Code :

- | | |
|----------------|-------------------|
| (a) 2, 3 and 4 | (b) 1, 2, 3 and 4 |
| (c) 3 and 4 | (d) 1 and 2 |

U.P.P.C.S. (Pre) 2017

I.A.S. (Pre) 1996

Ans. (a)

Idukki is not a thermal power project, but it is a hydroelectric power project which is situated across the Periyar River in Idukki district, Kerala. Sabarigiri hydroelectric project is in Kerala. Ghatprabha is an irrigation project and Ramganga is a multi-purpose project. Hence option (a) is correct.

3. Match List-I with List-II and select the correct answer using the codes given below the lists:

List- I

A. Ukai
B. Patratu
C. Pench
D. Dabhol

List- II

1. Jharkhand
2. Gujarat
3. Maharashtra
4. Madhya Pradesh

Code :

	A	B	C	D
(a)	4	2	3	1
(b)	2	1	4	3
(c)	3	2	1	4
(d)	1	3	4	2

U.P.P.C.S. (Pre) 2013

U.P. U.D.A./L.D.A. (Spl) (Pre) 2010

Ans. (b)

The correct match of thermal projects and their locations is as follows –

(Location)		(State)
Ukai	-	Gujarat
Patratu	-	Jharkhand
Pench	-	Madhya Pradesh
Dabhol	-	Maharashtra

4. Uran is a gas-based power project in –

- (a) Gujarat (b) Karnataka
(c) Maharashtra (d) Tamil Nadu

U.P.P.C.S. (Mains) 2010

Ans. (c)

Uran (Uran Gas Turbine Power Station) is a gas-based thermal power plant located in Raigad district of Maharashtra.

5. The largest source of energy in India is :

- (a) Nuclear (b) Petroleum
(c) Coal (d) Hydroelectricity

U.P. Lower Sub. (Pre) 2002

U.P.P.C.S. (Spl) (Pre) 2008

Ans. (c)

According to Power Ministry of Indian Government, India has electricity with a total installed capacity of 462001.66 MW on 31 December, 2024. In this thermal energy the highest share of 52.89% i.e. 244376.91 MW. Coal contributes 212349.50 MW, Lignite 6620.00 MW, Gas 24818.21 MW and Diesel 589.20 MW to thermal power installed capacity.

6. The highest share in power generation in India is of :

- (a) Atomic power
(b) Hydropower
(c) Thermal power
(d) All above have equal shares

U.P.P.C.S. (Mains) 2011

U.P.P.C.S. (Mains) 2006

Ans. (c)

See the explanation of above question.

7. The correct sequence of energy sources in order of their share in the power sector in India is :

- (a) Thermal > Hydro > Nuclear > Wind
(b) Thermal > Nuclear > Hydro > Wind
(c) Hydro > Nuclear > Thermal > Wind
(d) Nuclear > Hydro > Wind > Thermal

U.P.R.O./A.R.O. (Mains) 2014

U.P. U.D.A./L.D.A. (Pre) 2010

Ans. (*)

The correct sequence of energy sources in order of their share in the power sector in India is as follows:

As per the Ministry of Power, the installed power capacity in India as of December 31, 2024 was : Thermal (244376.91 MW), Wind (48163.16 MW), Hydro (46968.17 MW) and Nuclear (8180 MW). Thus, there is no proper sequence of energy sources obtained according to given options.

8. Consider the following means of energy production :

1. Water 2. Natural gas
3. Nuclear 4. Diesel

Arrange them in descending order according to their percentage (as on 30 September, 2019) of the total installed capacity of power generation in India.

- (a) 1, 3, 4, 2 (b) 3, 2, 1, 4
(c) 1, 2, 3, 4 (d) 2, 1, 4, 3

U.P.R.O./A.R.O. (Pre) 2016*

Ans. (c)

As on September 30, 2016, the percentage of total installed capacity of the means of energy production in India is as follows:

1. Water, 2. Natural Gas, 3. Nuclear, 4. Diesel.

Hence, it is clear among the given options, option (c) is the correct answer. It is notable that the total installed capacity in India at present (as on 31 December, 2024) is 462001.66 MW.

9. The superthermal electricity plant established by National Thermal Power Corporation (NTPC) in West Bengal is situated at :

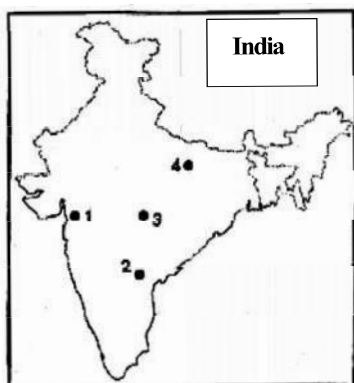
- (a) Asansol (b) Bairakpur
(c) Chandranagar (d) Farakka

R.A.S./R.T.S. (Pre) 1992

Ans. (d)

The superthermal electricity plant established by National Thermal Power Corporation (NTPC) in West Bengal is situated at Farakka. Farakka Super Thermal Power Plant is located at Nabarun in Murshidabad district in the Indian state of West Bengal. The power plant is one of the coal-based power plants of NTPC.

10. In the map given below, three out of the four places marked indicate places where a thermal power station is located:



The place where there is no thermal power station is marked:

- (a) 1 (b) 2
(c) 3 (d) 4

I.A.S. (Pre) 1995

Ans. (b)

The place marked '2' in the map is Srisailem where a hydroelectric power station is located in Telangana, not a thermal power station. The place marked '1' is Ukai thermal power station in Gujarat, '3' is Korba thermal power station in Chhattisgarh and '4' is Obra thermal power station in Uttar Pradesh. Hence (b) is the correct answer.

11. Match List-I with List-II and select the correct answer from the code given below :

List -I (Coal Based Power Plant)	List-II (Located in State)
A. Kahalgaon	1. Madhya Pradesh
B. Khargone	2. Bihar

C. Korba

D. Kudgi

Code :

- (a) A-1, B-2, C-4, D-3 (b) A-2, B-1, C-3, D-4
(c) A-1, B-2, C-3, D-4 (d) A-2, B-1, C-4, D-3

U.P. R.O./A.R.O. (Pre) 2023

Ans. (d)

Correct matched -

Coal Based Power Plant	Located in State
Kahalgaon	Bihar
Khargone	Madhya Pradesh
Korba	Chhattisgarh
Kudgi	Karnataka

12. The Neyveli thermal power plant is fed by :

- (a) Gondwana coal (b) Tertiary coal
(c) Quaternary coal (d) Cambrian coal

I.A.S. (Pre) 1993

Ans. (b)

The Neyveli thermal power plant is located in Tamil Nadu. It is the first lignite-fired thermal power station in Southeast Asia. Lignite is tertiary coal. Thus, option (b) is the correct answer.

13. Ramagundam Super Thermal Power Station is located in—

- (a) Andhra Pradesh (b) Jharkhand
(c) Madhya Pradesh (d) Karnataka

U.P. Lower Sub. (Spl) (Pre) 2004

Ans. (a)

Ramagundam Super Thermal Power Station, one of the biggest thermal power stations in India is situated in the then Andhra Pradesh (now Telangana). Its capacity is 2600 M.W.

14. With the help of which of the following countries, Thermal Power Centre at Obra was established?

- (a) America (b) Germany
(c) Japan (d) Russia

U.P.P.C.S. (Spl) (Mains) 2008

Ans. (d)

Obra Thermal Power Station is located in Sonbhadra district of Uttar Pradesh and was established with the help of Russian engineers in 1967.

15. Match List-I with List-II and select the correct answer using the codes given below the lists:

List- I	List- II
A. Kothagudem	1. Andhra Pradesh
B. Raichur	2. Gujarat
C. Mettur	3. Karnataka
D. Wankbori	4. Tamil Nadu

Code :

	A	B	C	D
(a)	4	2	1	3
(b)	1	3	4	2
(c)	4	3	1	2
(d)	1	2	4	3

I.A.S. (Pre) 2005

Ans. (b)

According to the given options, the correct match is as follows :

(Power station)	(State)
Kothagudem	- Andhra Pradesh (Now Telangana)
Raichur	- Karnataka
Mettur	- Tamil Nadu
Wankbori	- Gujarat

16. Which one of the following pairs is not correctly matched?

- (a) Badarpur - Delhi
(b) Harduaganj - Uttar Pradesh
(c) Utaran - Gujarat
(d) Paras - Andhra Pradesh

U.P.P.C.S. (Mains) 2006

Ans. (d)

According to given options, the correct match of thermal power plants and their locations is as follows :

Power Station	State
Badarpur	- Delhi
Harduaganj	- Uttar Pradesh
Utaran	- Gujarat
Paras	- Maharashtra

17. Match List-I with List-II and select the correct answer using the code given below the lists:

List- I (States)	List- II (Rank in installed thermal power capacity)
---------------------	--

- A. Gujarat 1. First
B. Maharashtra 2. Second
C. Uttar Pradesh 3. Third
D. West Bengal 4. Fourth

Code :

	A	B	C	D
(a)	4	3	1	2
(b)	2	1	4	3
(c)	3	4	1	2
(d)	1	2	4	3

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (*)

According to the given options, the correct match of rank in installed thermal power capacity and concerned States is as follows:

State power	Rank in installed thermal capacity as on 31.12.2024
Maharashtra	- First (28917.02 MW)
Gujarat	- Second (25145.93 MW)
Uttar Pradesh	- Third (23673.02 MW)
West Bengal	- Fourth (8763.34 MW)

18. Average All-India Thermal Plant Load Factor (PLF) in recent years has been varying between –

- (a) 60-65 percent (b) 65-70 percent
(c) 70-75 percent (d) 80-85 percent

U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (c)

During the time of this question, Average All India Thermal Plant Load Factor (PLF) had varied between 70-75%.

All India Thermal PLF Sector-wise (Excluding Gas Based Power Plants)		
Central	71.37	70.89
State	59.82	60.04
Private Sector IPP	64.37	66.16
Private Sector UTL	53.96	42.15
All India	65.07	65.43

19. The thermal power plant of Bokaro is located in :

- (a) Bihar (b) Chhattisgarh
(c) Jharkhand (d) Odisha

I.A.S. (Pre) 2003

Ans. (c)

Bokaro Thermal Power Station is located in Bokaro district in Jharkhand.

20. Which one of the following pairs is correctly matched?

- (a) Integrated Steel Plant at- Steel Authority of India Jajpur (Odisha)
(b) Power Plant at Jamnagar - Essar Power